

Blockchain and The Public Sector: Blockchain-Based Identity Management Systems For Public Services and The Impact on Privacy and Security Risks

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Abstract — This study explores the implementation of blockchain-based identity management in the Republic of Indonesia as well as analyse its impact on privacy and security risks. The methodology chosen for this research is a qualitative study using exploratory approach. The research propose six research propositions and a blockchain-based architecture model for citizen's personal identity management. This research finds that blockchain could mitigate privacy and security risks, should it be implemented by the Ministry of Home Affairs of the Republic of Indonesia.

Keywords — *blockchain, decentralized, distributed ledger, personal identity management, privacy risk, security risk*

I. Introduction

In the past, businesses were engaged in fierce competition to dominate valuable resources like oil and coal. However, the landscape has evolved significantly, transitioning from the industrial age to the digital age. In this new era, data has become the sought-after commodity, surpassing the significance of traditional resources [1]. The advent of computers and the Internet has led to the emergence of numerous technology companies striving to revolutionize the flow of data, aiming to enhance real-time information.

Ministry of Home Affairs of Republic of Indonesia states that for its citizen, they're still using their legacy internal server to store data of the citizen's personal identity, of which can be accessed by the local government [2]. This means the personal identity data is stored centrally within the Ministry as provider and can be accessed by the local government through cloud. This generate the risk of security and privacy violations considering that citizen's personal identity data is a sensitive information. Citizen as user must carefully evaluate the risk of privacy breaches since privacy is considered a fundamental right of the user. Any violation of this right by the provider would signify unethical behaviour on their part [3][4].

According to CNBC Indonesia, there are more than 200 million Indonesian citizen's personal identity that are sold in the dark web [5]. The personal identity data are hacked in 2021 and sold in Bitcoin. Furthermore, CNN Indonesia also mentioned that there are more than 100 million Indonesian citizen's personal identity that are leaked during 2022 [6]. As the personal identity contains the name, phone number, address, and e-mail address, the leaked data significantly

harm the users, or in this case, the citizen since the data could be used for phishing, unauthorized profiling, or any other unethical action by the person who owns the data. This leads to a privacy and security risks that must be addressed by the society as well as the Ministry of Home Affairs of Republic of Indonesia [7].

Enter blockchain, a potential solution to these issues. Blockchain is a fully distributed system that effectively captures and stores a consistent and sequential record of transactions among interconnected participants within the ecosystem [8]. Its decentralized nature allows blockchain to lower transaction costs and provide tamper-proof evidence of transactions [9].

It's important to note that adopting blockchain requires substantial computing capabilities, resulting in financial considerations for organizations due to the need for high-end computing infrastructure [10]. Despite its potential, blockchain is still in its early stages and faces technical challenges [11]. However, it is worth noting the contrasting nature of blockchain's decentralization and the centralized nature of cloud computing. The aim of this research is to explore the possible implementation of blockchain on citizen's personal identity management as well as investigate the impact on security and privacy issue.

II. Literature Review

Nakamoto's paper introduced bitcoin as a peer-to-peer electronic cash system that enables direct online payments between parties without relying on intermediaries like financial institutions within a specific network. Since bitcoin is not a recognized currency, it requires a ledger mechanism that ensures transaction security, and that's where blockchain comes into play. Blockchain can be described as a distributed ledger that maintains a continuously growing list of confirmed data records by the network participants, known as nodes [12].

When a transaction occurs in the bitcoin network, an individual uses their private key to generate a binary code signature. This signature can be verified by other

network participants using the person's public key. The bitcoin is then transferred to another person, along with a hash value that is automatically generated based on the previous person's information, as depicted in Figure 1. The hash value, comprising numbers and symbols, is created using a cryptographic hash function. Modifying the hash value deliberately would require guessing the input used to generate it. Even a slight difference in the input would completely change the hash value, making it extremely difficult to tamper with.

Once the transaction is agreed upon by the involved nodes, it is recorded and placed in a block, as shown in Figure 1. Each block in the blockchain has its own unique hash value and is linked to the previous and subsequent blocks, forming a complete ledger of transaction history [13]. Since blockchain records transactions sequentially, each block must refer to the previous block to be considered valid. This design prevents tampering with the ledger, as any alteration of a single element within a block would affect the entire transaction history within the network.

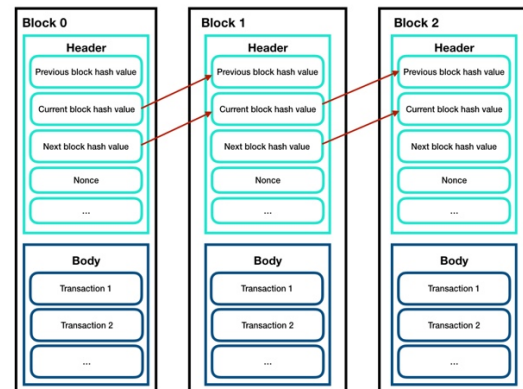


Figure 1. Blockchain Mechanism.

Smart contracts are a recent technology that has gained attention, originally introduced by Nick Szabo in 1997. They are computerized protocols that execute contract terms by translating them into code. This code is embedded into hardware or software, enabling self-enforcement of contractual clauses. Combining blockchain with smart contracts allows for the transfer of assets between parties who don't trust each other. It also enables multi-step procedures to occur among mutually distrustful parties. This broader application of blockchain technology facilitates secure and

autonomous processes among multiple nodes in a network. The concept of blockchain can be adopted into other organisational context since it would reduce transaction costs and manages resources on the outside as easy as managing the resources on the inside of an organisation [14].

III. Methodology

Exploratory studies aim to investigate phenomena that are not well understood and generate ideas for future research [15]. These studies typically focus on answering "what" and "how" questions [16]. Given the limited resources and objectives of this study that are mentioned in the previous chapter, an exploratory approach appears the most appropriate.

The data collected for this research will primarily be secondary data due to limitations in time and resources. Secondary data refers to data that were originally collected for other purposes [16]. The sources of data for this research may include, but are not limited to, organizational databases, reports, books, journal articles, magazine articles, newspaper articles, interview transcripts, press releases, and conference proceedings that are relevant to cloud computing and blockchain. Given that the topic is still at its infancy stage, the search term will be limited to 'blockchain', 'decentralize', 'distributed ledger', 'public sector', 'identity management', 'security', and 'ethical risks'. The data collected then analysed with three steps. First, extract important and relevant passages from the data; second, define the meaning of extracted texts and/or passages and label it into codes and concept; and third, find meaningful and similar pattern as well as relationship in between codes.

IV. Results and Discussion

Based on the data collected, there is pattern that indicates blockchain's dimension as described on Table 1.

Table 1. Blockchain Dimensions.

Dimension	Sub-Dimension	Author(s)
Cost	Initial Investment	(Battleson et al. 2016); (Halaburda 2018); (Joshi 2017); (Podder et al. 2017); (Singh and Michels 2018); (Tosh et al. 2017); (Xu et al. 2017)

	Time	(Lu and Xu 2017); (Zheng et al. 2018)
Flexibility	Formation	(IBM 2018); (Maull et al. 2017); (Priyadarshinee et al. 2017)
	Provider Involvement	(Li et al. 2018); (Manski 2017); (Scott et al. 2017); (Zhang et al. 2017)
	Mechanism	(Gatteschi et al. 2018); (Shermin 2017)
Security	Data Integrity	(Huang et al. 2017); (Kewell et al. 2017); (Nagaraj and Maguire 2017); (Steininger 2018) (Xia et al. 2017); (Yermack 2017); (Zhang et al. 2018);
	Transparency	(Baxendale 2016); (Hsieh and Wu 2018); (Jacobides et al. 2018); (Seidel 2018)
	Regulations	(Herian 2017); (McConaghy et al. 2017); (Yeoh 2017); (Zhang et al. 2017)
Scalability	-	(Liang et al. 2017); (Maull et al. 2017); (Renner et al. 2018); (Scott et al. 2017);
Performance	-	(Botta et al. 2016); (Lu and Xu 2017) (Tosh et al. 2017); (Yang et al. 2018)

Cost

Cost pertains to the resources that customers must allocate for the implementation, maintenance, and enhancement of technology, whether it is cloud computing or blockchain. From the analyzed data, the most common cost pattern observed is the initial investment, such as the upfront investment in blockchain technology. Due to its decentralized nature, users interested in utilizing blockchain must invest in computing hardware of at least above-average quality. Enterprises, in particular, need to allocate significant resources to acquire hardware infrastructure for storage and computational power [17].

Organizations considering the adoption of blockchain technology must take into account the need for significant investments in computing hardware. Unlike cloud computing, blockchain technology requires more time for data processing, demanding high computational power from enterprise users. While the public sector may not face budget constraints, the adoption of blockchain still requires substantial financial investment and additional time for data processing.

Research Proposition 1 (RP1): Blockchain initial investment is costly to be implemented for citizen's personal identity management

Flexibility

Flexibility refers to the level of customization available in blockchain. Blockchain has a certain deployment model that allows enterprise users to implement the technology according to their specific

business process requirements. This flexibility is also what enables blockchain to be utilized beyond cryptocurrency [18]. Despite access limitations in blockchain technology, it provides users with a similar degree of flexibility. This has led to the emergence of blockchain as a service (BaaS) offered by cloud providers like IBM and Microsoft, who collaborate with Hyperledger and Ethereum, respectively. This emergence of BaaS would enhance the degree of flexibility in implementing blockchain, regardless of the business process. However, the use of BaaS would also endanger the citizen's data since the application of BaaS would mean that the government entrusts the data to the BaaS provider's server.

Research Proposition 2 (RP2): Blockchain is flexible enough to be implemented for citizen's personal identity management

Security

Alongside cost, security emerged as another frequent pattern during the data analysis. The security dimension refers to the level of protection against malicious attacks. As data is a valuable asset in the information age, security is a major concern. Codes embedded in the blockchain system can introduce vulnerabilities that make the data susceptible to malicious attacks [19]. Additionally, the decentralized nature of blockchain necessitates a certain degree of transparency to ensure the immutability of data, but this transparency may be perceived as a breach of privacy by some users [20]. Security has been a major concern holding back the adoption of cloud computing. The pooling of customer data in one or more data centers makes it easier for hackers to attack, as breaching one data center puts all the users' data at risk. This is particularly concerning for users who store sensitive information in the Ministry of Home Affairs' data center.

The hype around blockchain suggests that it could replace cloud computing due to its highly secure mechanism. The degree of security in blockchain depends on the deployment model. Public blockchains offer high transparency, enhancing security but raising privacy concerns. Tracing data provenance through blockchain architecture has been proposed as a solution, allowing auditors to track data usage and

preventing improper data use by cloud providers [21][22]. This model benefits enterprise users by leveraging the transparency provided by the decentralized nature of blockchain technology. Permissioned blockchain models restrict transaction visibility to specific nodes, enabling users to manage the level of transparency by also adding a node specifically to monitor the flow of data within the blockchain ecosystem known as the blockchain architecture auditor. However, security concerns persist, as blockchain is not entirely immune to attacks such as the 51% attack or collusion between nodes and providers/auditors. Currently, The Ministry of Communication and Informatics of the Republic of Indonesia had published one regulation concerning the implementation of blockchain [23]. The lack of regulation regarding blockchain technology in Indonesia generates concern whether it would be beneficial to adopt the technology. On the other hand, this lack of regulatory framework concerning blockchain could also be beneficial for the adopter as a premature regulation by the government could hinder the growth of the technology itself.

Research Proposition 3 (RP3): Blockchain will improve the security aspect on citizen's personal identity management

Scalability

Scalability focuses on the technology's ability to adapt to users' needs. Blockchain technology faces challenges in adjusting to varying demands. For instance, when an enterprise wants to join a consortium blockchain ecosystem, the network must reconsider the consensus mechanism, requiring deliberation among the ecosystem's nodes [24]. The nature of the blockchain mechanism, where transactions are hashed and added to blocks sequentially, introduces complexity and time-consuming processes. This may not be suitable for data that requires frequent movement. Moreover, the dynamic nature of the personal identity environment, with frequent changes in nodes (i.e. adding more and more citizen within the blockchain ecosystem), adds complexity to blockchain implementation. Changing nodes intermittently affects network rules and imposes additional workload for maintaining data integrity,

which becomes a concern when implementing blockchain for citizen's personal identity management.

Research Proposition 4 (RP4): Blockchain will require massive amount of time and effort to implement on citizen's personal identity management

Performance

Performance refers to the efficiency of task execution by the technology. Blockchain technology demands high processing power from users since data is not stored in a centralized database. Users of blockchain must invest in hardware to generate sufficient processing power for storing and transferring data, utilizing the capabilities of the hardware equipment. As the blocks in blockchain grow in size, proper computing hardware is necessary to ensure effective performance. Should the Ministry decides to adopt blockchain, it must invest in computational power to enable the nodes responsible for maintaining the data (miners) to perform effectively, regardless of block size. Even though one could argue that the government has no budget constraint, the computing performance alone cannot be considered as the main aspect in effectiveness of performance. The stakeholder within the blockchain ecosystem must work accordingly in order to achieve the effective implementation of blockchain in personal identity management. This would require efforts and time invested by the nodes within the blockchain ecosystem.

Research Proposition 5 (RP5): Blockchain has high computational power requirement if to be implemented on citizen's personal identity management

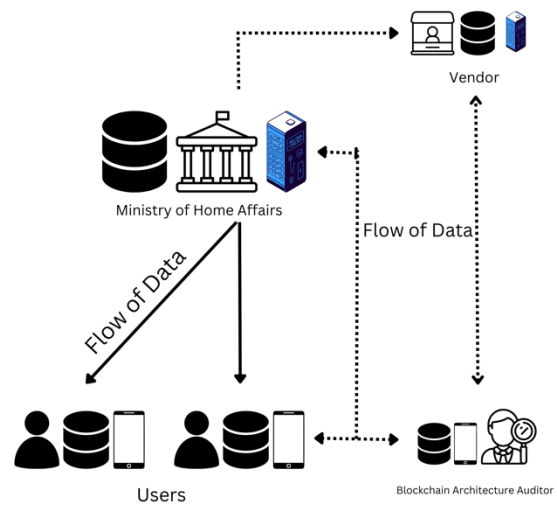


Figure 2. Possible Future Integration of Blockchain on Personal Identity Management

Figure 2 describes the possible future implementation of blockchain on personal identity management. As it can be seen from Figure 2, the stakeholder (i.e. the nodes) in the ecosystem are the government, the users (i.e., citizen), the vendor, and the blockchain architecture auditor. Each users within the ecosystem would always know should there be any movement of data concerning their personal identity. Figure 2 also suggests that implementing blockchain on personal identity management could enhance transparency for users to track their data. Furthermore, the blockchain mechanism offers robust security, making it challenging for any party to tamper with the data. By deploying blockchain in personal identity management along with the existence of blockchain architecture auditor, the nodes would be compelled to act responsibly, potentially mitigating the privacy concerns associated with personal identity management and thus addressing privacy issues.

Research Proposition 6 (RP6): Blockchain could mitigate security and privacy issue on citizen's personal identity management

V. Conclusions

This research had explored the implementation of blockchain on personal identity management and its impact on privacy and security issue by collecting and analysing relevant secondary data. Scholars have been

conducting experiments in regards to the implementation of blockchain in many sectors. The experiments conducted by scholars indicate that blockchain could improve security and transparency in the future. Therefore, if the blockchain mechanism deployed, the nodes would have to behave appropriately since the security and transparency mechanism would put the users in a stronger position as the users could monitor the movement of their data, which consequently could mitigate ethical problem such as privacy issue. Thus, we recommend that the Ministry of Home Affairs of Republic of Indonesia implement blockchain technology on personal identity management. However, there seems to be no indication that the government moving towards this deployment which might be due to the costly initial investment as well as the immaturity of blockchain technology in the Republic of Indonesia.

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